Chapter 1
Introduction

Objectives

- Data / Information / Knowledge / Wisdom
- What is a database?
- What does database design do?
- What is database management system (DBMS)?

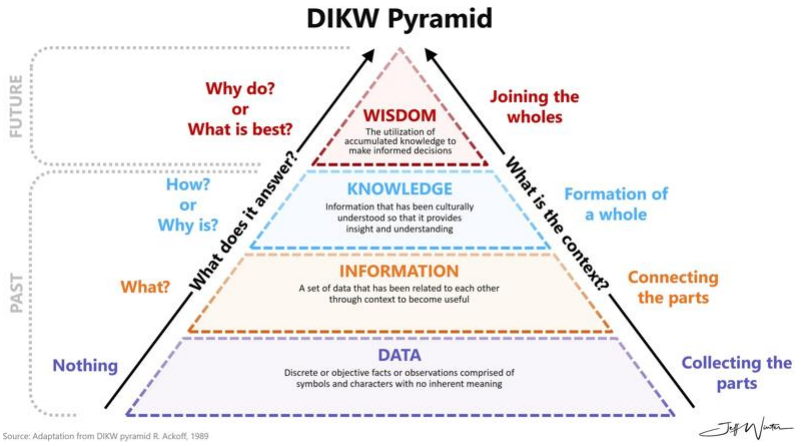
Data / Information

- Data: recording of known facts about separate objects or events in applications.
  - Using numbers and strings, certain features of the objects or events are described and stored on a storage medium.
- Information is the semantics conveyed by data. It represents the meaningful context of data items.
  - For example, collecting a patient's data including name, address, date of birth, body weight etc. to represent health information of an individual.

Knowledge / Wisdom

- Knowledge provides an understanding of information.
  - For example, given a patient name and address, from the date of birth and body weight, we may conclude that the patient may be overweight for the relevant age group
- Wisdom – knowledge supports informed decisions.
  - For example, patient records may strongly suggest that smoking is a major cause of lung cancer.
  - This knowledge will help in driving future actions to improve the habits of patients to reduce the risk of the disease.

DIKW Pyramid



Why the need to store data?

- Data enable an organization to keep records about events that occur.
- Information enables the organization to react and respond to the events.
- Knowledge enables the organization to anticipate events and act appropriately when the events occur.

Database

- A database is a single, large, organized collection of data that can be used simultaneously by many departments and users.
- It is the collection of data that contains information relevant to an enterprise.
- Databases allow data to be stored quickly and easily and are used in many aspects of our daily life.
- Your school, grocery store, bank, and clothing store all use databases to keep track of customer, inventory, employee and accounting information.

Types of Database

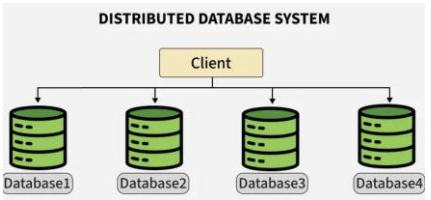
- **Relational Databases:**
  - Data stored in structured tables with predefined schemas
  - Using SQL (Structured Query Language) for data definition, manipulation, and querying
  - Examples of Relational Database Management System (RDBMS)
    - MySQL, PostgreSQL, SQLite, MariaDB, Oracle, etc.

Types of Database

- NoSQL (Not Only SQL) Databases:
  - Handles large volumes of unstructured or semi-structured data, and rapidly changing data
  - Flexible and scalable
  - Prioritizing scalability and performance over strict data consistency
  - Type: Document, Key-value, wide-column, graph, etc.
  - Examples: MongoDB, Redis, etc.

Types of Database

- Distributed databases
  - Data stored across different physical locations, connected via a network, managed as a single logical database
  - High availability, fault tolerance
  - Examples of DDBMS: Amazon DynamoDB, Apache Cassandra, etc.



Database Design

- **Database design** is the organisation of data according to a database model.
- The designer determines what data must be stored and how the data elements interrelate.
- In the case of relational model, data are represented in the form of tables. Each table has multiple columns, and each column has a unique name. Each row of the table represents one piece of information.

Database Design – Relational Model

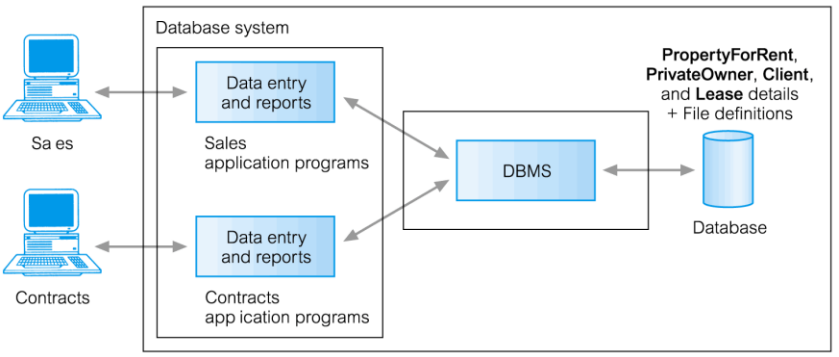
Additional tasks for Relational Model:

- ER diagram (entity-relationship model)
- Normalization
  - *normalization* is a systematic way of ensuring that a database structure is suitable for general-purpose querying and free of certain undesirable characteristics — insertion, update, and deletion anomalies that could lead to loss of data integrity.

Database Management System (DBMS)

- A database management system (DBMS) provides the software tools needed to organize that data in a flexible manner.
- Handles concurrency control to manage simultaneous transactions without corrupting data.
- It includes tools to add, modify or delete data from the database, ask questions (or queries) about the data stored in the database and produce reports summarizing selected contents.
  - In RDBMS, we use SQL (Structured Query Language) statements to manipulate the data

Database Processing (Fig 1.7)



PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)
Lease (leaseNo, propertyNo, clientNo, paymentMethod, deposit, paid, rentStart, rentF nish)

Summary

We have covered the following:

- **Difference between a database and Database Management System (DBMS)**
- Purpose of database design
- Examples of RDBMS